Discounted Cash Flow (DCF) Valuation — handout

Lesson Forge Pro (owner)

Lesson plan

Discounted Cash Flow (DCF) Valuation

Bloom's levels: Apply, Evaluate

Objectives

- Apply DCF valuation techniques to estimate business value
- Calculate cost of capital (WACC) and discount rates
- Evaluate the strengths and limitations of DCF valuation

Prerequisites

- Financial Statement Forecasting and Analysis

Key points

Discounted Cash Flow (DCF) Valuation

Session Overview

In this 120-minute session, we will explore the concept of Discounted Cash Flow (DCF) valuation, its application in estimating business value, and the calculation of cost of capital (WACC) and discount rates. By the end of this session, you will be able to apply DCF valuation techniques, calculate cost of capital, and evaluate the strengths and limitations of DCF valuation.

Timed Agenda

Time	Topic
0:00 - 5:00	Introduction to DCF Valuation
5:00 - 15:00	Main Elements of DCF Valuation
15:00 - 25:00	Calculating Cost of Capital (WACC) and Discount Rates
25:00 - 35:00	Energizer/Break
35:00 - 50:00	Applying DCF Valuation Techniques
50:00 - 65:00	Evaluating the Strengths and Limitations of DCF Valuation
65:00 - 80:00	Case Study: DCF Valuation in Practice
80:00 - 120:00	Group Discussion and Q&A

Introduction to DCF Valuation (0:00 - 5:00)

Welcome to our session on DCF valuation! To start, let's consider an analogy. Imagine you have a friend who promises to pay you $100 in one year. However, you need the money now, so you ask your friend to give you the present value of that $100. To calculate this, you would use a discount rate, which reflects the time value of money. This is similar to how DCF valuation works, where we estimate the present value of future cash flows using a discount rate.

Main Elements of DCF Valuation (5:00 - 15:00)

The main elements of DCF valuation are:

Free Cash Flow Projections: Projections of the amount of cash produced by a company's business operations after paying for operating expenses and capital expenditures.

Discount Rate: The cost of capital (Debt and Equity) for the business.

Terminal Value: The value of a business at the end of the projection period.

Let's use an analogy to illustrate these concepts. Think of a company as a tree that produces fruit (cash flows) over time. The free cash flow projections represent the expected fruit production, the discount rate represents the cost of waiting for the fruit to grow, and the terminal value represents the value of the tree after the projection period.

Calculating Cost of Capital (WACC) and Discount Rates (15:00 - 25:00)

To calculate the cost of capital, we use the Weighted Average Cost of Capital (WACC) formula:

WACC = (Cost of Debt x Debt %) + (Cost of Equity x Equity %)

The discount rate is then used to calculate the present value of future cash flows. For example, if we have a cash flow of $100 in one year, and a discount rate of 10%, the present value would be:

PV = $100 / (1 + 0.10)^1 = $90.91

Energizer/Break (25:00 - 35:00)

Let's take a short break to stretch and refresh our minds. During this time, consider the following question: What are some common challenges in estimating future cash flows, and how can we address them?

Applying DCF Valuation Techniques (35:00 - 50:00)

Now, let's apply DCF valuation techniques to a sample company. We will estimate the free cash flow projections, calculate the cost of capital, and determine the terminal value. Using these inputs, we can calculate the present value of the company's future cash flows and estimate its intrinsic value.

Evaluating the Strengths and Limitations of DCF Valuation (50:00 - 65:00)

DCF valuation has several strengths, including:

It takes into account the time value of money

It provides a framework for estimating the present value of future cash flows

However, DCF valuation also has some limitations, including:

It requires accurate estimates of future cash flows

It is sensitive to changes in discount rates and growth assumptions

Let's discuss these strengths and limitations in more detail, and consider the following question: How can we address the limitations of DCF valuation in practice?

Case Study: DCF Valuation in Practice (65:00 - 80:00)

Let's work through a case study to illustrate the application of DCF valuation in practice. We will estimate the intrinsic value of a company using DCF valuation, and discuss the results.

Group Discussion and Q&A (80:00 - 120:00)

Now, let's open the floor for a group discussion and Q&A. Some anticipated questions and answers include:

Q: What is the difference between DCF valuation and accounting book value?

A: DCF valuation estimates the present value of future cash flows, while accounting book value represents the historical cost of assets.

Q: How do we estimate the cost of capital?

A: We can estimate the cost of capital using the WACC formula, which takes into account the cost of debt and equity.

Q: What are some common challenges in applying DCF valuation in practice?

A: Some common challenges include estimating future cash flows, determining the discount rate, and addressing the limitations of DCF valuation.

I hope this session has provided a comprehensive overview of DCF valuation and its application in practice. Thank you for your participation, and I look forward to your questions and comments!